HARDIK SRIVASTAVA

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EDUCATION

University of Washington — Seattle, WA
Master of Science in Data Science (GPA - 3.9/4) — Sept 2025 - Mar 2027

SRM Institute of Science and Technology — Chennai, TN
Bachelor of Technology in Computer Science with specialization in Big Data Analytics (GPA - 3.82/4) — May 2019 - Jun 2023

WORK EXPERIENCE

Applied Scientist Intern — June 2026 - Present
JPMorganChase — *Seattle, WA*

- Architected a billion-scale Temporal Graph-ML system for **adversarial actor modeling**. Optimized **context utilization** by building a **sequence-compression encoder** fusing recent-k events with 2-hop summaries and scaled multi-GPU training via a **C++ sampler**.
- Designed an inductive framework using time-bucketed **contrastive ranking** and logQ debiasing to neutralize popularity bias, boosting relative precision by **9.2%**; engineered self-supervised event reconstruction with dynamic memory-dropout for robust zero-shot inference.

Graduate Research Assistant (P.I.: Prof. Zaid Harchaoui) — Sept 2025 - Present
Paul G. Allen School of Computer Science, University of Washington — *Seattle, WA*

- Architected a **post-training optimization** system for authorship obfuscation by fine-tuning a multi-adapter LoRA ensemble (Qwen2.5-32B) for dynamic style transfer; built metric-driven agents to guide **alignment** via constrained decoding, improving privacy by **40%**.
- Productionized the **agentic workflow** on **HPC clusters** utilizing Mistral-24B for robust evaluation and policy refinement. Engineered multi-objective feedback loops to enable **real-time policy iteration**, maintaining **88%** semantic similarity and **5%** perplexity drift.

Applied Scientist — Jun 2023 - Aug 2025
JPMorganChase — *Hyderabad, TS*

- Proposed a custom **cost-sensitive Focal Loss objective** to penalize false negatives in payment fraud detection; retrained production models to secure a **6%** recall boost and **$220M+** in annual savings while utilizing TreeSHAP to maintain strict feature interpretability.
- Designed an end-to-end ML system using a **multimodal Transformer** (ViT+BERT) for OCR check fraud detection, boosting precision by 19% over baselines. Optimized inference via **mixed-precision** to handle 500K+ daily requests with minimal p99 latency.
- Built a **Learning-to-Rank information retrieval** system over **3B+** KYC records using Pairwise Ranking loss to flag anomalous entities; cut **62% false positives** outperforming the TF-IDF baselines, and saved **700+** analyst review hours daily in production.
- Engineered **distributed ETL pipelines** in AWS Glue for **terabyte-scale ML training**. Developed **automated feature selection** workflows that boosted downstream fraud capture by **14%**, minimizing manual engineering overhead and reducing time-to-deployment.

Research Intern — Jun 2022 - Sept 2022
McGill University - *Mitacs Scholar* — *Montreal, QC*

- Engineered a DeBERTa-based **dense retrieval** system with **FAISS** for predictive vocabulary generation, driving a **6.9% MAP** gain.

Machine Learning Engineer Intern — Jan 2022 - Jun 2022
Samsung Research — *Bangalore, KN*

- Optimized production-scale **NLU intent routing** for Samsung Bixby via neural graph matching, reducing misrouted intents by **12%**; Deployed INT8 inference via **Quantization-Aware** training for word disambiguation models, surpassing FP32 baselines by **2.9%** F1.

PROJECTS

Recursive Episodic Memory RAG: [link] Engineered a **memory-compression** RAG framework with FAISS hierarchical retrieval, semantic clustering, and relevance ranking to enable **persistent multi-session reasoning** for long-context LLM agents. Achieved **41% context compression** under strict token budgets by implementing recursive episodic memory consolidation via SLMs and adaptive replay.

PokerFace-RL [link] Built a multi-agent reinforcement learning framework using PPO and QLoRA to train LLMs in **strategic deception** during asymmetric-information dialogue. Developed a two-stage **opponent-masked** reasoning architecture, modeling interactions as a **POMDP** with GAE, within a custom Gymnasium environment, increasing deception success by **13pp** over zero-shot baselines.

Agentic Ad Intelligence Engine [link] Architected a **multi-agent orchestration** pipeline for X (Twitter) using **Grok 4.1** to generate personalized ad copy for 100+ dynamic user personas, achieving **sub-5-second** end-to-end latency. Engineered a dynamic routing mechanism that ranked and delivered advertisements based on live engagement feedback, boosting click-through rates by **16%** in pilot campaigns.

SELECTED PUBLICATIONS

Srivastava, H. (2023) Multi-Modal Sentiment Analysis Using Text & Audio for Customer Support Centers. *(ICACTCE, Springer Nature)*

Srivastava, H. (2023) Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems. *(ICKES, IEEE)*

SKILLS

CS/ML C++, Python, Java, SQL, PyTorch, TensorFlow, Sklearn, Spark, CUDA, LangGraph, HuggingFace, Generative AI

Infra/MLOps AWS, Kubernetes, LangChain, MLFLow, RAG, A/B Testing, Distributed Systems, Feature Stores, Vector Search